

① Esame 05/02/2024 (NON È DETTO SIA TUTTO GIUSTO)

① Tabella di verità di F in AB

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

② $F = AB\bar{C}$

③ $Y = \overline{(AB\bar{C} \cdot C)} = 1$

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1.

library ieee

use ieee.std-logic-1164.all

entity Azzurro is

port (A, B, S: in std-logic

C: out std-logic)

end Azzurro

entity Rosso is

port (S, B, C: in std-logic

Y: out std-logic)

end Rosso

entity Verde is

port (A, B, S: in std-logic

Y: out std-logic)

end Verde

2. library ieee

use ieee.std_logic_1164.all

architecture Behaviour of AZZURRO is
begin

if S = '0' then

C <= not A

else

C <= (A and B) or (not A)

end Behaviour

architecture Structure of Verde is

signal C : std_logic

begin

AZZURRO : AZZURRO port map (A, B, S, C)

ROSSO : ROSSO port map (S, B, C, Y)

end Structure

③ Tabella di verità

x	y	z	L	D _x	D _y	D _z
0	0	0	0	0	0	1
0	0	0	1	1	0	0
0	0	1	0	0	0	1
0	0	1	1	1	0	0
0	1	0	0	1	1	1
0	1	0	1	1	1	1
0	1	1	0	1	1	1
0	1	1	1	1	1	1
1	0	0	0	0	0	1
1	0	0	1	1	1	0
1	0	1	0	1	1	1
1	0	1	1	1	1	1
1	1	0	0	0	0	0
1	1	0	1	1	1	0
1	1	1	0	1	1	1
1	1	1	1	1	1	1

2) Exp minima di D_x

$\begin{smallmatrix} x/y \\ \hline zL \end{smallmatrix}$	00	01	11	10
00	0	-	0	0
01	1	-	1	1
11	1	-	-	-
10	0	-	-	-

$$D_x = L$$

Exp minima di D_y

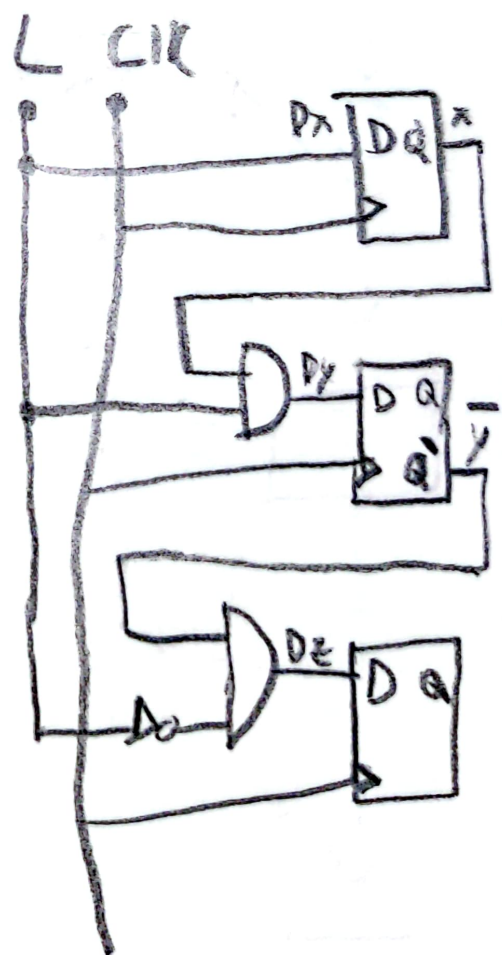
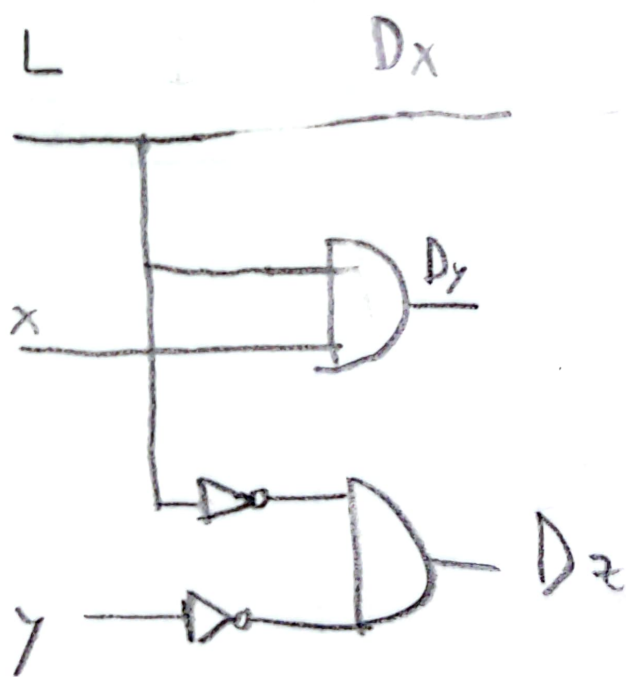
$\begin{smallmatrix} x/y \\ \hline zL \end{smallmatrix}$	00	01	11	10
00	0	-	0	0
01	0	-	1	1
11	0	-	-	-
10	0	-	-	-

$$D_y = L_x$$

Exp minima di Dz

xy	00	01	11	10
z				
00	1	-	0	1
01	0	-	0	0
11	0	-	-	-
10	1	-	-	1

$$Dz = \bar{y} \bar{z}$$



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